

Md Takiuddin

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Senior Backend Engineer with 6+ years building resilient, observable microservices and event-driven systems. Strong across Python (FastAPI), TypeScript (NestJS), Go, and Rust, with a bias toward clean API contracts and observability from day one. Ships production systems for globally distributed teams and mentors engineers through code review and pairing.

PROFESSIONAL EXPERIENCE

COGNITUS, AN IBM COMPANY

Senior Software Developer

Remote

November 2023–Present

- Core back-end contributor on LambdaX – an enterprise contract-lifecycle-management platform used by large US customers
- Designed and shipped agreement-header, line-item, dataset, and modification microservices using Python (FastAPI), SQLAlchemy, and PostgreSQL
- Extended the in-house low-code flow orchestrator with reusable action blocks spanning SAP, Outlook, DocuSign, and Teams
- Built a Redis Streams + Kafka event fabric and a Socket.io-to-Elasticsearch audit pipeline aligned with SOC-2 retention
- Building performance-critical services in Rust for event processing and latency-sensitive paths
- Tuned slow SQLAlchemy queries, introduced Prometheus alerting, and mentored junior engineers through code review
- Built an MCP server with FastMCP and an in-app chatbot using PydanticAI with pgvector-backed retrieval

GRYPE DIGITAL

Software Developer

Remote

May 2022–November 2023

- Led the back-end squad for Member Lounge, a SaaS engagement platform for associations and non-profits (NestJS, TypeScript, MongoDB, RabbitMQ)
- Architected tiered subscription billing on Stripe with proration, coupons, webhook processing, and automated invoicing
- Shipped a threaded discussion forum and a queue-driven notification engine across browser push, email, and SMS
- Rolled out MFA (email OTP + trusted-device cookies) and zero-downtime blue/green deploys on Docker Swarm
- Solo-built the HPSA portal on FastAPI and Next.js, working directly with the CEO on specs and demos

BUGFIXERS

Full-Stack Developer

Dhaka, Bangladesh

March 2020–April 2022

- Delivered products across Node.js, Python, React, Flutter, and Electron for clients including LearnGenie, Devsfit, and the Bangladesh Navy
- Built secure offline-capable intranet desktop apps and real-time Socket.io dashboards for operational awareness
- Introduced RabbitMQ + Redis-backed microservice decomposition, GitHub Actions CI/CD, and hardened Linux VPS deployments
- Acted as first responder for production incidents and refactored legacy PHP/Vanilla-JS into modern Express + React stacks

KEY PROJECTS

PROCHESTA – AI EXAM-PREP PLATFORM

CTO & Engineering Lead

Independent Project

2025–Present

- Bangladesh's leading job-exam prep platform (BCS, bank, government) with model tests and performance analytics
- Built an AI tutor and Bangla AI study assistant with pgvector-backed retrieval
- Shipped offline-ready Flutter apps alongside a Next.js web app
- Own the architecture, CI/CD, observability, and performance across Node.js, MongoDB, and FastAPI services

TAXSTAR – UAE CORPORATE-TAX SAAS

Solo Architect & Developer

Independent Project

August 2023–February 2024

- Built end-to-end FinTech SaaS helping UAE businesses comply with federal corporate-tax rules
- Implemented rules-driven engine for automated tax calculations, forecasting, and deadline reminders
- Integrated with Xero and Zoho ERP systems to eliminate manual CSV uploads
- Designed multi-tenant architecture ready for horizontal scaling and future VAT/payroll-tax modules

HONEST ELITE – SOCIAL-MEDIA PLATFORM

Full-Stack Developer

Independent Project

September 2022–March 2024

- Developed rich-media social network with real-time chat, voice/video calls, and personalized recommendations
- Built GraphQL API on NestJS microservices with Next.js SSR frontend and WebRTC for calls
- Implemented a collaborative and content-based recommender system to improve feed relevancy and engagement
- Used Redis Streams for real-time messaging and call notifications

TECHNICAL SKILLS

- **Languages:** Python, TypeScript/JavaScript, Go, Rust, SQL, Bash
- **Backend & APIs:** FastAPI, NestJS, Flask, Django, Express, gRPC, GraphQL
- **Datastores:** PostgreSQL, MongoDB, Redis, MySQL, Elasticsearch, pgvector
- **Messaging & Streaming:** Apache Kafka, RabbitMQ, Redis Streams, Socket.io
- **Cloud & DevOps:** Docker, Kubernetes, AWS, Cloudflare, NGINX, GitHub Actions, Linux
- **Security & Observability:** Keycloak, OAuth 2.0/OIDC, JWT, MFA, Prometheus, Grafana, OpenTelemetry
- **AI & Data:** LangChain, LangGraph, PydanticAI, FastMCP, scikit-learn, pandas
- **Frontend:** React, Next.js, Tailwind CSS, Flutter

EDUCATION

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

Bachelor of Science in Mechanical Engineering

Rajshahi, Bangladesh

2016–2022

- Capstone: "Low-Cost Autonomous Line-Follower Robot for Warehouse Logistics" – 1st runner-up in RUET TechFest 2019
- Technical Secretary, Robotic Society of RUET; software lead for Team Scrapper (Indian Rover Challenge)
- Coursework: Thermodynamics, Fluid Mechanics, CAD/CAM, Control Systems, Materials Science, Industrial Automation

LANGUAGES

- Bangla (Native), English (Full Professional), Hindi (Limited Working)